Kaustubh Gavankar

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Summary

Data science student passionate about software development and machine learning. Seeking opportunities to apply technical and analytical skills to real-world problems, contribute to impactful projects and grow as a developer.

Technical Skills

- Languages: Python, SQL, C++, HTML,CSS
- ML/DL: Scikit-learn, TensorFlow, PyTorch, XGBoost
- LLM & NLP: Hugging Face Transformers, Sentence Transformers, LangChain, RAG, LoRA, PEFT
- Data Analysis: Data Visualization, EDA, Feature Engineering, Statistical Analysis
- **D**atabases: PostgreSQL , MySQL , Vector DB
- Tools: PowerBi , Tableau, Excel, Jira.
- Data Skills : ETL (SQL , Python-Pandas) , Data Cleaning

Experience

National Remote Sensing Centre (NRSC), ISRO Jan 2026 – july 2026

- Worked as an **AI/ML Intern** at **NRSC (ISRO)**, contributing to projects focused on intelligent information access and retrieval systems.
- Build a **Document Retrieval System** for NRSC's internal website, aimed at efficiently searching and retrieving information from large, structured datasets.
- Designed and implemented text preprocessing pipelines, including data cleaning, tokenization, and vectorization for improved search accuracy.

Bharat Space Education Research Centre (BSERC) July 2026 – August 2026

- Developed AI agents capable of performing autonomous, multi-step task execution using Large Language Models.
- Build agentic workflows integrating reasoning, planning, tool calling, and decision-making to solve real-world tasks.
- Implemented Retrieval-Augmented Generation pipelines using vector embeddings and semantic search for context-aware responses.

Projects

Sentinel-1 SAR-Based Land Use/Land Cover Classification {ISRO-PROJECT 2}

- Designed and implemented an **end-to-end SAR image classification pipeline** for automated Land Use/Land Cover mapping using **Sentinel-1** satellite imagery.
- Extracted and engineered **VV/VH radar backscatter features**, applying SAR preprocessing and speckle filtering using **Google Earth Engine** and Python.
- Trained and evaluated **Machine Learning models** for multi-class LULC classification, generating high-resolution thematic maps with quantitative performance evaluation.

Emotion Detection Using EEG (CNN-based Model) github.com/KaustubhGavankar/emotion-detection-cnn

- Built a Convolutional Neural Network (CNN) model to classify human emotions using EEG signal data.
- Preprocessed time-series data and implemented model training using TensorFlow and Keras.

AlpaCare Medical Instruction Assistant github.com/KaustubhGavankar/alfacarefolder

- Fine-tuned a <7B parameter pre-trained LLM using the lavita/AlpaCare-MedInstruct-52k dataset to develop a safe, non-diagnostic medical instruction assistant.
- Implemented parameter-efficient fine-tuning (LoRA/PEFT) on Google Colab with GPU optimization .
- Tech Stack: Python, PyTorch, Hugging Face Transformers, PEFT/LoRA, Pandas, NumPy.

Bayesian Network for Weather Prediction github.com/KaustubhGavankar/Bayesian-Network-for-Weather-Prediction

- Developed a probabilistic weather forecasting system using Bayesian Networks to model dependencies between atmospheric variables.
- Performed data preprocessing, feature selection, and conditional probability modeling to improve prediction reliability.
- Implemented the solution in Python, leveraging probabilistic inference for interpretable weather predictions.

Education

- Bachelor of Technology in Data Science Engineering
  GH Raisoni College Of Engineering And Management , Nagpur {7.1 CGPA} Graduation: 2026
- Class XII | HSC | MSBSHSE
  Major Hemant jakate College Of Science and Commerce ,Nagpur {77%} March 2022
- Class X | SSC | CBSE March 2020
  Bhavans B.P Vidhya Mandir ,Civil Lines ,Nagpur {80.2%}

Certifications

- Python Programming Internship Certificate – CodSoft (July 2024)
- AI/ML internship NRSC- ISRO (july2026)
- AI Internship BSERC (August2026)
- Data Analysis with AI/ML NareshIT (ongoing)